

States of matter

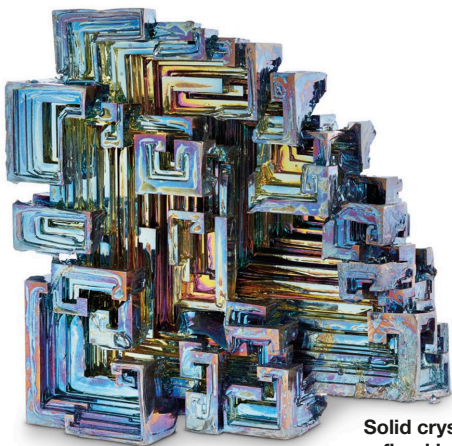
Every element has a standard state – solid, liquid, or gas – at room temperature. The atoms of a solid fit together in a tight-knit pattern, while in a liquid the atoms are loosely connected so they flow around. In a gas, the atoms are free of each other and disperse easily. Applying heat can change the state of an element from solid to liquid, and then to gas, or even from solid to gas in some cases.



Chlorine gas in a glass sphere



Liquid mercury in a vial



Solid crystals of bismuth refined in a laboratory

Pure forms

A pure sample of an element contains only atoms of that element. Only a few elements are found pure in nature in significant amounts. These include gold and sulfur (in the ground), and oxygen (in the air). Many elements are found in ores.

Gold in quartz



This vein of pure gold has occurred naturally inside a chunk of quartz.